

Writing of reports/documentation

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Documentation:

- Text
- Computations
- Drawings
- Figures
- Graphs
- Pictures
- Diagrams

Communication between two parties - the author communicates a message to the reader.

Structure of a report

- **You** are the author(s) = **you** decide on the structure
- Think carefully on the situation of the reader
 - The reader does not know what you know
 - In what order does the reader need the information to understand the report?
- How can you help the reader?

Usually this structure **does not work**

Preliminaries	Front page/title page Abstract (Foreword) List of content (Nomenclature)
Main text	Introduction Theory Methods Results Discussion Conclusion
End matters	References/literature Enclosures Appendices

Structure more suited for mechatronics

Preliminaries	Front page/title page Abstract (Foreword) List of content (Nomenclature)
Main text	- Introduction and state-of-the-art - Concept generation, selection, presentation - Presentation of final design (basis for the rest of the report) - Control calculations (document that the design will work correctly - mechanics, hydraulics, simulations etc.) - Results - Discussion - Conclusion
End matters	References/literature Enclosures Appendices

Structure of a Report

- A report is not written in chronological order
- The structure of a report does not reflect the order it is written but the order it is read.
- A good structure helps the reader

Formalities

- Standardized title page?
- Page setup?
- Font and font size?
- Colors or black/white?
- Page limitations?
- References?
- Writing of equations?
- Drawing material
- Language

Page setup, font and font size

- Not too small margins
- Not too small line spacing
- Font size 10-12 for the body text
- On paper a font with serifs (**feets on the letters**). Look in the text books.
- Rarely lower than 3 levels in the list of content (Chapter > Section > Subsection)

General: make it easy for the reader.

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Colors or black/white

- Use colors if it increases the understanding of the material, - else not
- Do not use colors just to make it fancy
- Consider the printing process (your self)
- The author is responsible that colors can be read/seen

General: make it easy for the reader.

References

- In the text and the reference list/bibliography
- My recommendation: Harvard APA
- www.referencing.port.ac.uk/apa/index.html
- <http://uia.no/kildebruk>
- Footnotes
 - The reader must move the eyes.
 - My point of view: if it is important then write it in the text, - if not then leave it out.

General: make it easy for the reader.

References

List of references/bibliography:

Last name, Initials. (Year). *Title*. Place: Publisher. Maybe. ISBN.

Arora, J.S. (1989), *Introduction to optimum design*, 1st ed., New York: McGraw-Hill Primis Custom Publishing.

Stephens, R.I., A. Fatemi, R.R. Stephens and H.O. Fuchs (2001), *Metal Fatigue in Engineering* 2nd ed., New York: John Wiley & Sons, Inc.

The method of the conjugated gradient is an optimization routine of first order [Arora, 1989, maybe page number]. Therefore it is applied in...

References

List of references/bibliography:

Author, *Title in italic*. Place: Publisher, Year.

- [1] J. Keats, *Virtual words: language on the edge of science and technology*. Oxford: Oxford University Press, 2011.
- [2] M.-B. O. Nielsen and J. Thon, *Wergelandenes Kristiansand: bygguide*. Kristiansand: Wergelandselskapet, 2008.

The method of the conjugated gradient is an optimization routine of first order, see [\[1\]](#). Therefore it is applied in...

References

- Equation numbers
- Figure numbers
- Table numbers
- Chapter numbers/section numbers
- Appendices

Consecutive numbers in the report or in each chapter.

Refer to the numbers. More precise than "above", "below", "earlier", "previous page" etc.

Drawings

- Scale (1:1, 1:2, 1:5, 1:10)
- Scaled versus paper size
- Line width
- Informations
- Folder for drawings
- Be careful about "shrink to fit"when printing

Language

- Choose a language and use the correct words
- Solid vs. massive
- Stress vs. spenning
- Strain vs. tøyning
- Force, moment, kraft, moment
- Velocity, speed, angular velocity, angular speed (one does not compute the rpm)

Writing

- Grammar.
- Spelling (Spell check) 😊.
- Correct suffix, singular/plural, present tense/past tense, perfect tense/pluperfect tense (førnutid/førdatid).
- Punctuation.

Writing

- Correct terminology
- Write short, clear, and precise.
- Write the message on the line, - not between the lines.
- Be careful about writing too private.
- The number of pages is not a quality index.
- Quotation marks
 - He is a "clever" boy.

AI Text Generation

If/when you use text tools:

- Read the text **very carefully AND correct it when wrong**
- Is the content correct?
- Is the correct terminology used - correct words?
- Is the language appropriate?
- Are the references correct?
- Often AI does NOT supply the necessary figures

AI Text Generation

I asked for an explanation of free-body-diagrams

Example: Free Body Diagram of a Simple Beam

Consider a beam supported at both ends, with a load applied in the middle:

1. The beam is represented by a straight line.
2. Forces acting on the beam:
 - A vertical applied force ($F_{applied}$) is acting downward at the center of the beam.
 - The beam has two supports: a pin support at one end and a roller support at the other end.
 - The pin support exerts two reaction forces: a vertical force ($R_{vertical}$) and a horizontal force ($R_{horizontal}$).
 - The roller support exerts only a vertical reaction force ($R_{vertical}$).
3. Moments:
 - If the beam is subjected to bending, there may be moments at the supports depending on how the forces are applied.
4. ...

Avoid Nonsense (AI or human)

Example #1

The lift cylinder is designed with a foot mounting installation to provide secure base fixation, improved stability, and efficient load path alignment within the lifting mechanism.

Avoid Nonsense (AI or human)

Example #2

In the tilt hydraulic circuit, counterbalance valves are also installed to prevent uncontrolled forward tilting of the mast caused by the load moment and to ensure smooth and stable tilting motion.

Avoid Nonsense (AI or human)

Example #3

The hydraulic actuation system is based on the load-sensing (LS) control principle. By automatically adjusting the pump output pressure and flow rate, the system maintains efficient, stable, and safe operation under various load conditions.

Who is the reader?

Who?

- The president
- Head of department
- Colleague
- Fellow students
- Supervisor/examiner
- Family
- ...

Background and needs?

- Knowledge about the topic
 - In the group
 - Outside the group
- Mathematical knowledge
- No prerequisites
- Time available
- ...

The interests of the author

It is the interest of the author that the message is understood.



The author must:

- take the reader's knowledge into account
- help the reader during the reading
- explain the material at an appropriate level
- use an appropriate language (try to avoid: I/we did...)
- ...

How to help the reader?

Explain at a basic level what you are working with.

REMEMBER that the reader has not worked with it as intensively as you have

"What is a dump truck/dumper?"

How to help the reader?



Figur 1: <http://www.volvo.com/dealers/no-no/Volvo/newsmedia/image+gallery/introduction.htm>

How to help the reader?



Figur 2: Private picture.

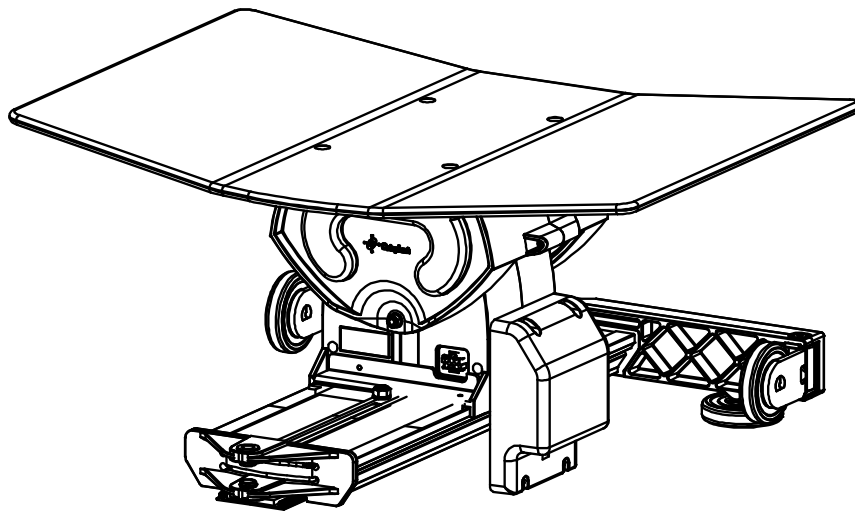
How to help the reader?



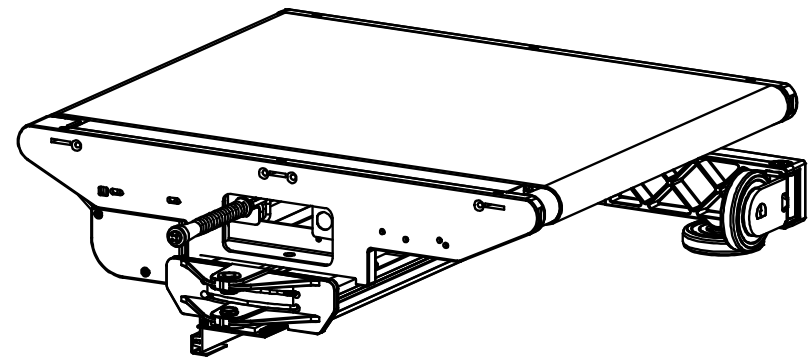
Figur 3: Picture from Thomas H. Langer and Morten K. Bak.

How to help the reader?

Explain the structure:



Tilt unit

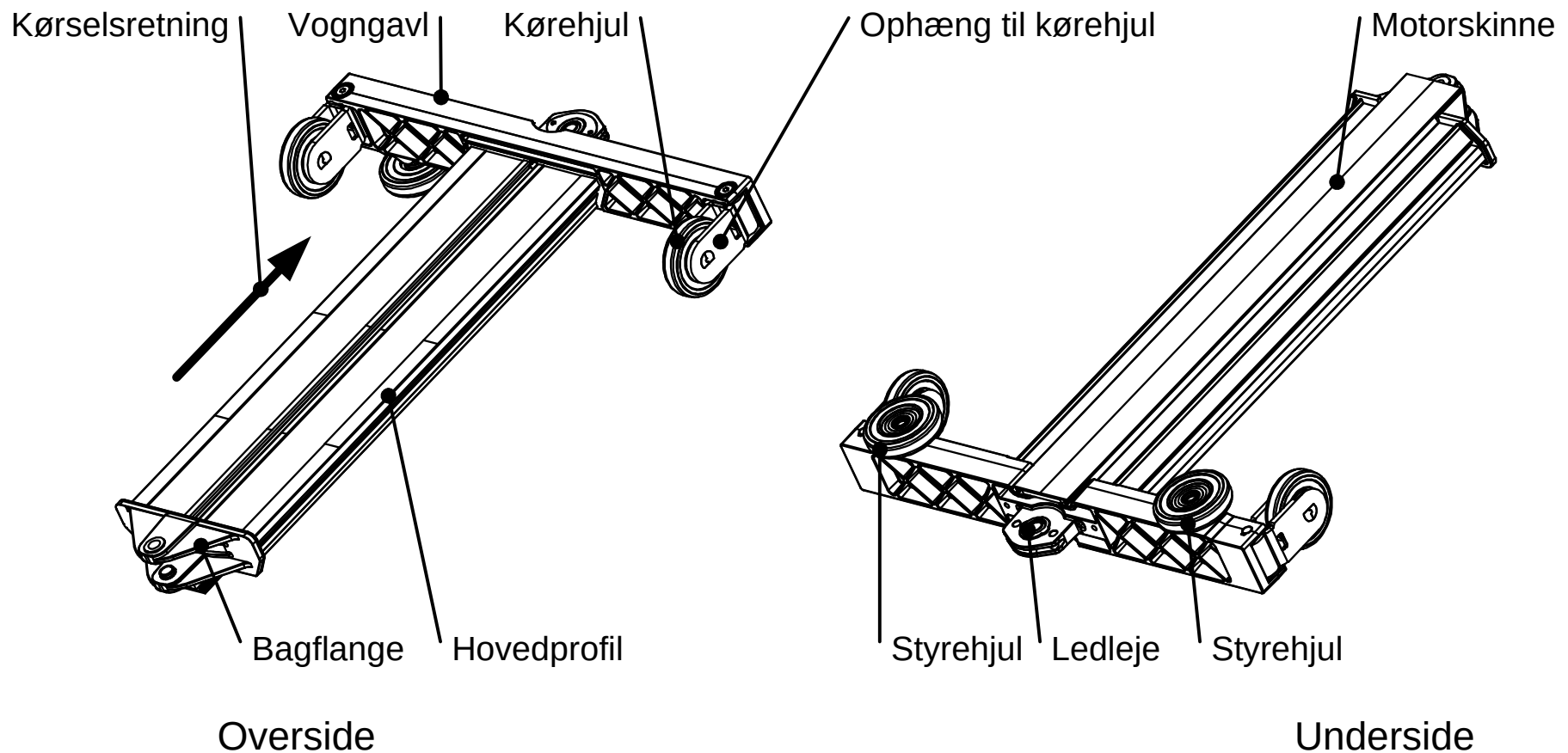


Cross belt

Figure 4: Cart with tilt unit or cross belt.

How to help the reader?

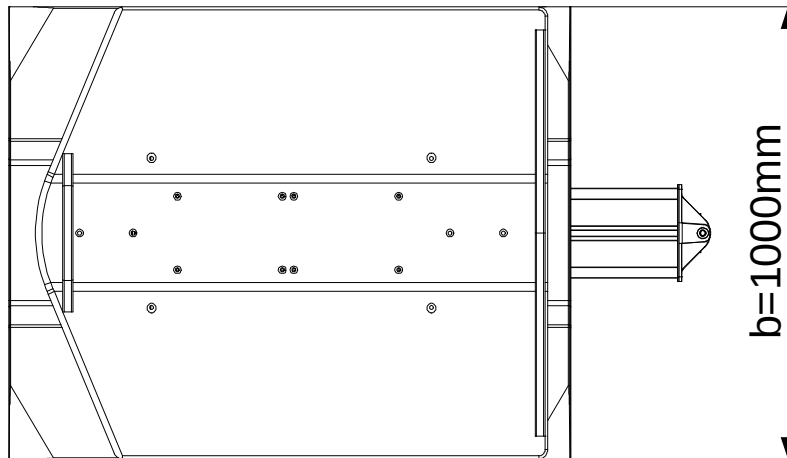
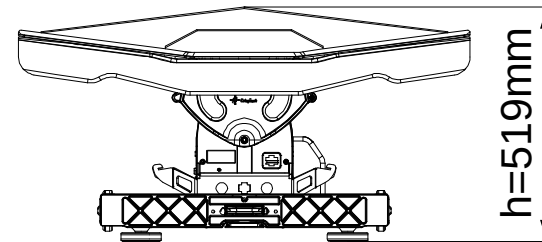
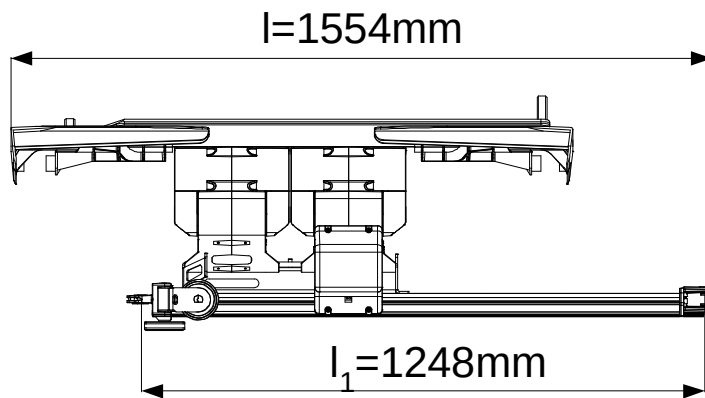
Explain the structure (if necessary a foldout):



Figur 5: 750mm cart with designations.

How to help the reader?

Show the main dimensions:



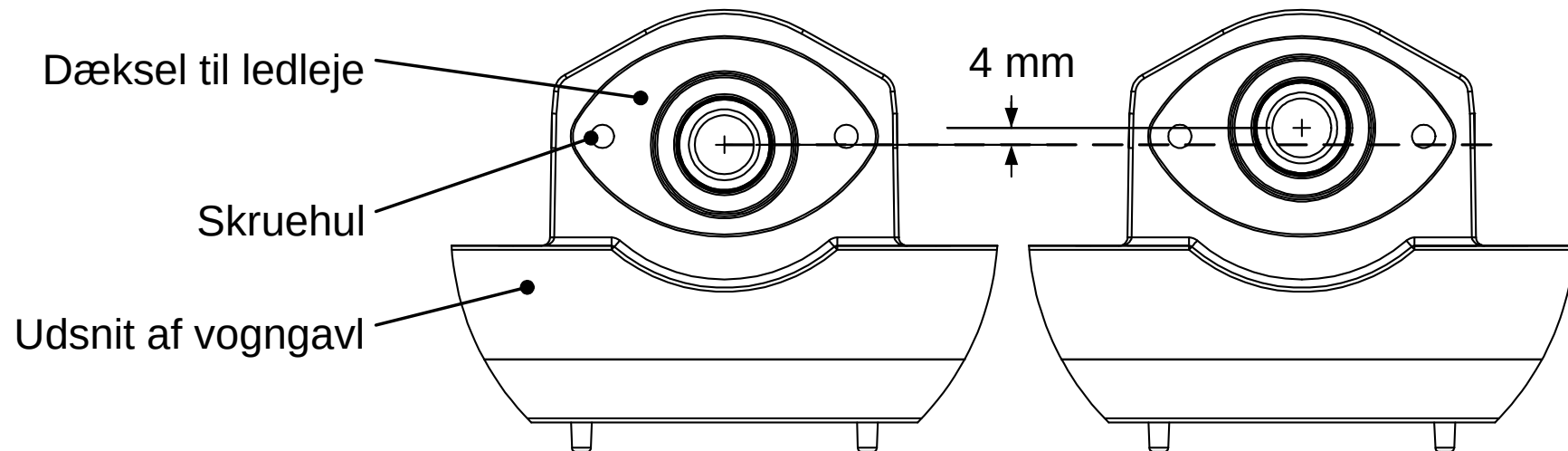
Figur 6: 1248mm cart with main dimensions.

How to help the reader?

The length of the cart can be changed 4mm with the aid of a ball joint.

How to help the reader?

The length of the cart can be changed 4mm with the aid of a ball joint.



Figur 7: Excentric ball joint.

How to help the reader?

The cart is modeled with springs to the surroundings in all the contact points.

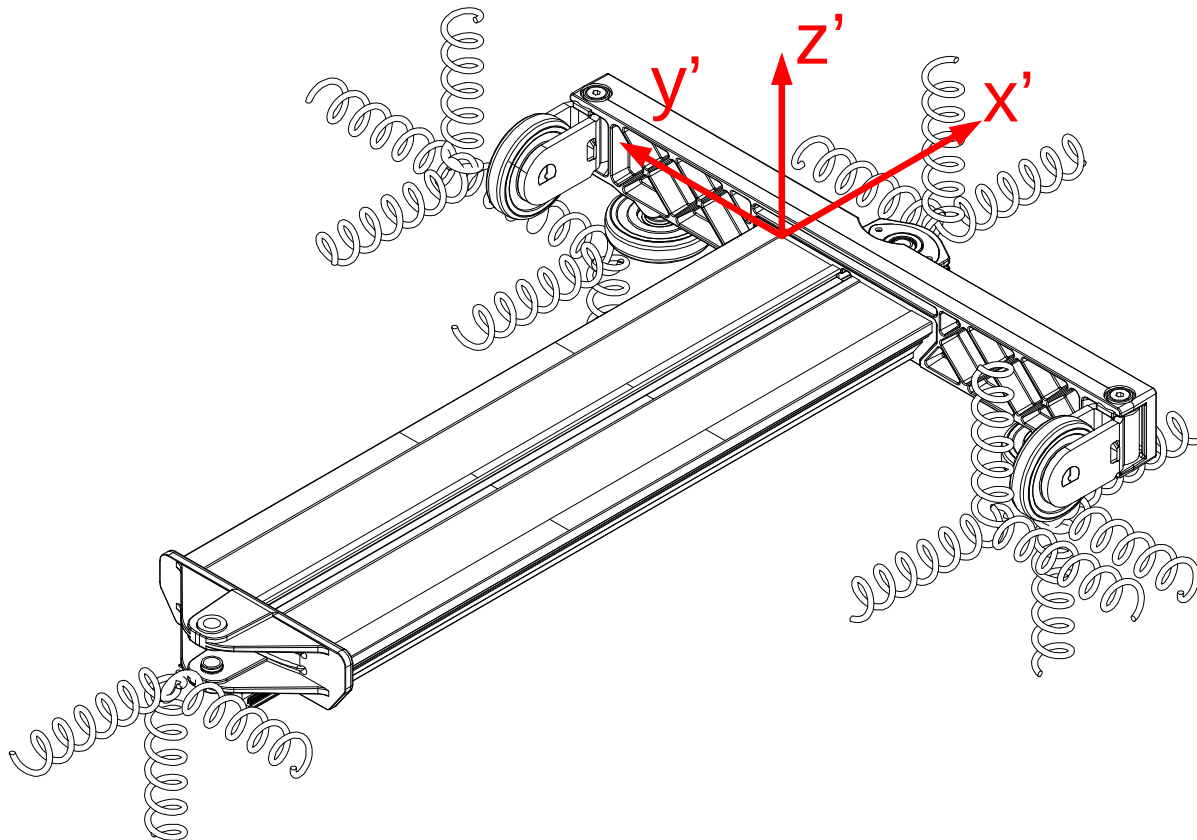
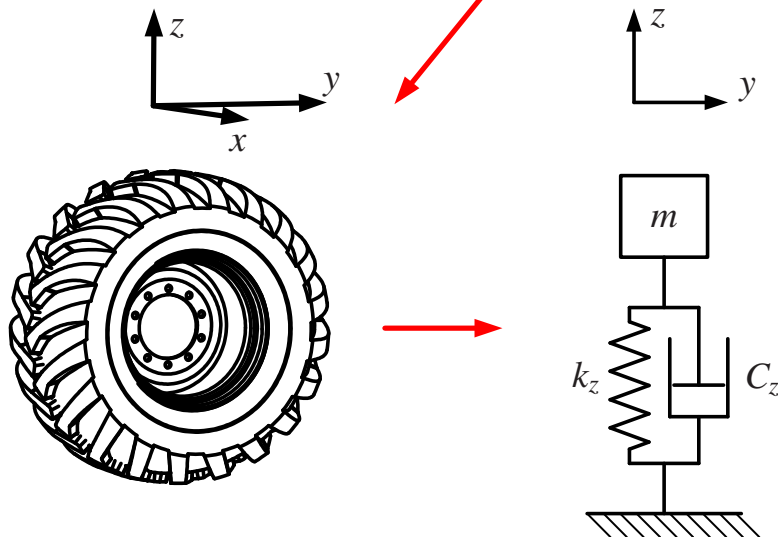


Figure 8: Cart with springs in the contact points.

How to help the reader?

"The wheel of the dump truck is a linear spring and damper."



Reality $>$ $<$ models

- Assumptions
- Prerequisites
- Limitations
- Linear spring and damper
- ...

"The wheel of the dump truck is modeled as a linear spring and damper with the characteristics..."

Computations - FBD

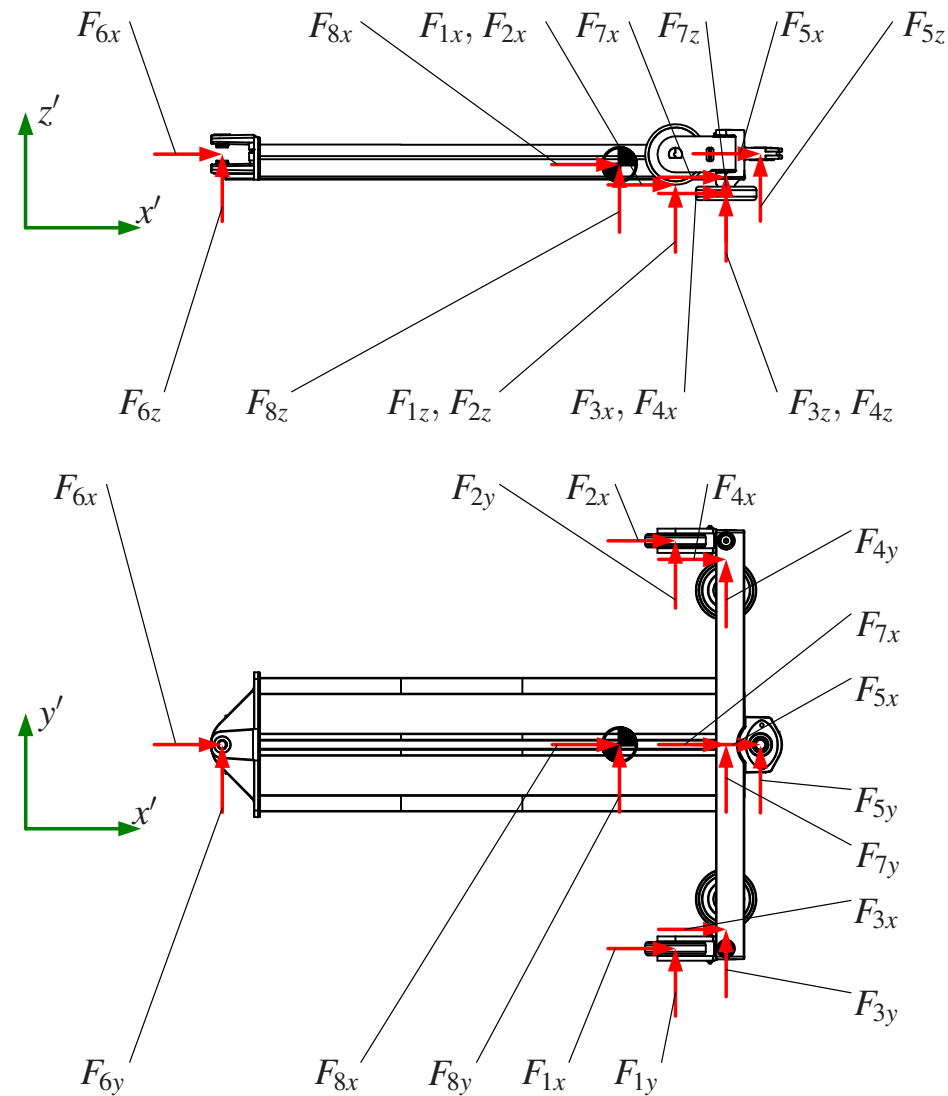
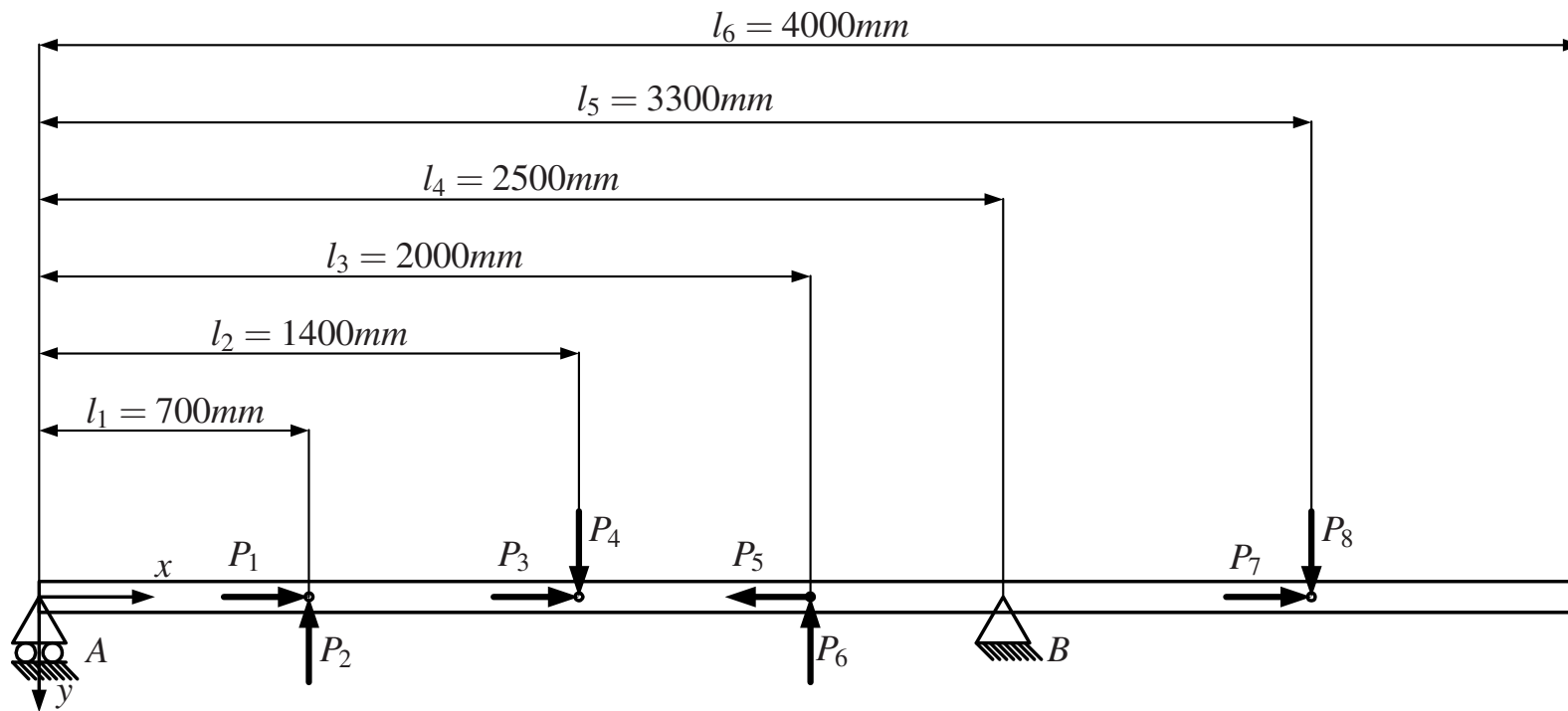
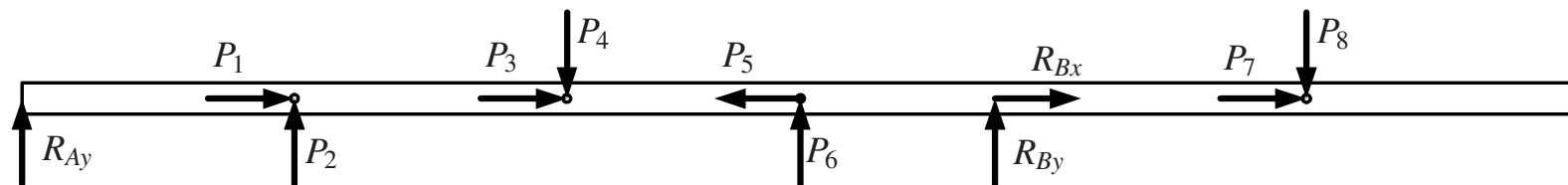


Figure 9: Free body diagram for the cart in the dynamic model.

Computations - FBD



Figur 10: Simply supported beam.



Figur 11: Free body diagram of the beam shown in figure 10.

Computations

- Nomenclature (inform the reader about it)
- Clear notations
 - Upper case/lower case letters
 - Bold/not bold
 - Italic/not italic
 - Vectors/scalars
 - F , $\mathbf{F}$, $\vec{F}$, f
- Same formatting of symbols on figures, in the text, and in equations
- Correct designations (force, mass, pressure, flow)
- Units ($[\text{N}]$, $[\text{kg}]$, $[\frac{\text{N}}{\text{m}^2}]$, $[\frac{\text{m}^3}{\text{s}}]$)

15. Analyse av usikkerhet

Du skal analysere produktet med hensyn til om noen viktige suksesskriterier er usikre:

Vi har undersøkt hvilke belastningene plastkomprimatoren må tåle med vår trykkløsning og hvilke vanntrykk som skal være i norske husstander, og dette skal ligge på mellom min 3 og Max 11 bar, Vi har i tillegg lagt til en sikkerhetsmargin på 1,3 ved.

15.1 Tykkelse:

Rørtykkelse sylindere 6 mm, platetykkelse 6 mm.

15.2 Diameter:

Sylinder ytre mål 300 mm – indre mål 296 mm
Stempel ytre mål 296 mm – indre mål 288 med mer

15.3 Max trykk:

Max trykk i sylindren ved 3 bar (300000 Pa)

$$F = P * A$$

$$F = ((\pi * 0,3^2 \text{ m} / 4) * 300000 \text{ Pa}) : 9,81 \text{ m/s}^2 = 2161 \text{ Kg}$$

Max trykk i sylindren ved 11 bar (1100000 Pa)

$$F = P * A$$

$$F = ((\pi * 0,3^2 \text{ m} / 4) * 1100000 \text{ Pa}) : 9,81 \text{ m/s}^2 = 7926 \text{ Kg}$$

Max trykk i sylindren ved 11 bar (1100000 Pa) med sikkerhetsmargin på 1,3

$$F = P * A$$

$$F = ((\pi * 0,3^2 \text{ m} / 4) * 1100000 \text{ Pa}) : 9,81 \text{ m/s}^2 * 1,3 = 10303 \text{ Kg}$$

15.4 Belastninger:

De største belastningene i konstruksjonen vil komme i boltene som holder lokket på plass, i sveisen til lokket og i bunnen flensens sveising til sylindren. For å styrke disse delene har vi forsterket lokket og bunnen med tykkere plater.

Naming of variables

General rule: one letter/symbol in normal size and the rest in subscript (or superscript). There are only few well established exceptions:

- Reynolds number: Re
- Froude number: Fr
- Mach number: Ma
- Prandtl number: Pr
- Péclet number: Pe
- Darcy number: Da
- Sherwood number: Sh
- Rayleigh number: Ra
- Grashof number: Gr
- Nusselt number: Nu
- Biot number: Bi
- Fourier number: Fo
- Stanton number: St
- Knudsen number: Kn
- Eckert number: Ec

There may be other exceptions

Do not invent new ones

Naming of variables

General rule: one letter/symbol in normal size and the rest in subscript (or superscript).

$$\overset{\text{wrong}}{\text{Area}} = \overset{\text{wrong}}{\text{radius}} \cdot \pi$$

$$A = r \cdot \pi \quad \text{correct}$$

$$\overset{\text{wrong}}{\text{Crossectionarea}} = \overset{\text{wrong}}{L_4} \cdot \overset{\text{wrong}}{L_3}$$

$$A_{\text{cross}} = L_4 \cdot L_3 \quad \text{correct}$$

$$\overset{\text{wrong}}{rpm} = \frac{v}{r}$$

$$\omega = \frac{v}{r} \quad \text{or} \quad \dot{\theta} = \frac{v}{r} \quad \text{correct}$$

Multiplication sign

Multiplication signs can be included or left out.

As a minimum be consistent and use the same sign for the entire report. Possible signs:

- Small dot: $\cdot$ (`\cdot`)
- Large dot: $\bullet$ (`\bullet`)
- Cross: $\times$ (`\times`)

Personally I recommend a small dot to avoid misunderstandings when leaving out the sign.

Mathematical functions

It is recommended to use the build in functions to emphasize when a mathematical function is being used (both in Word and Latex):

<i>wrong</i>	<i>right</i>	
$\sin a$	$\sin a$	(\sin)
<i>wrong</i>	<i>right</i>	
$\cos a$	$\cos a$	(\cos)
<i>wrong</i>	<i>right</i>	
$\tan a$	$\tan a$	(\tan)
<i>wrong</i>	<i>right</i>	
$\log a$	$\log a$	(\log)
<i>wrong</i>	<i>right</i>	
$\ln a$	$\ln a$	(\ln)

Computations

One computation or a set of computations

Purpose



Theoretical background



Equation or equations

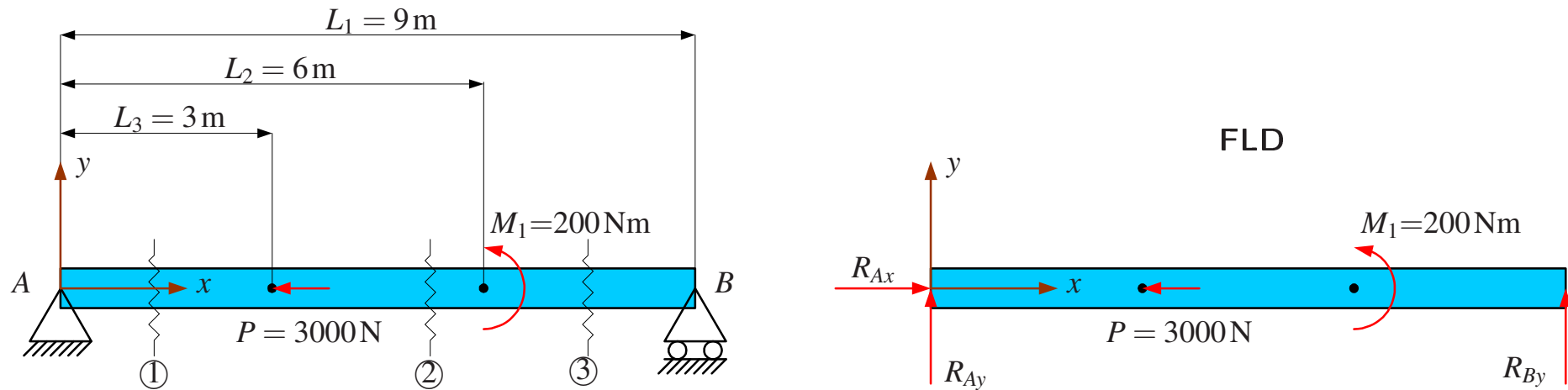


Presentation of the result (one number, table, graph etc.)



Conclusion

Computations



Figur 12: Simply supported beam with an axial force and a moment and the associated free-body-diagram FBD.

$$\begin{aligned}\sum \overset{\oplus}{M}_A &= 0\text{ Nm} = M_1 + \begin{bmatrix} L_1 \\ 0\text{ m} \end{bmatrix} \times \begin{bmatrix} 0\text{ N} \\ R_{By} \end{bmatrix} \\ &= M_1 + L_1 \cdot R_{By} - 0\text{ m} \cdot 0\text{ N}\end{aligned}\quad (1)$$

$$\sum F_x = 0\text{ N} = R_{Ax} - P \quad (2)$$

$$\sum F_y = 0\text{ N} = R_{Ay} + R_{By} \quad (3)$$

A lot of algebra may not be relevant but the initial equation and the solution are important.

Computations

Symbolic expression → show the correct numbers → result

$$\begin{aligned} E_{kin} &= \frac{1}{2} \cdot m \cdot v^2 \\ &= \frac{1}{2} \cdot 23 \text{ kg} \cdot (12.5 \frac{\text{m}}{\text{s}})^2 \\ &= 1796.875 \text{ J} \end{aligned} \tag{4}$$

or

$$E_{kin} = \frac{1}{2} \cdot m \cdot v^2 = \frac{1}{2} \cdot 23 \text{ kg} \cdot (12.5 \frac{\text{m}}{\text{s}})^2 = 1796.875 \text{ J} \tag{5}$$

Explain the numbers (You ask for this yourself in the lectures)

Remember units (Check the units - often you can find errors this way)

Computations

- Show the equation with symbols.
- Solve the equation symbolic.
- Maybe just show the solution without the algebra in between.
- Insert numbers **with units**.
- Show the result.
- Conclude on the result.

You cannot leave it to the reader to conclude if the result is acceptable or not. You are the engineer = you should do the job you are asked to do.

Computations

- Be careful about copying directly from MathCad, Maple, Matlab or other programs.
- Remember the explanation to the equation (Where and what).
- Do not present many equations without text. The reader has no chance to understand it.
- Accuracy of the result? (Significant digits).

Exercise OØM-MKE-18 solution

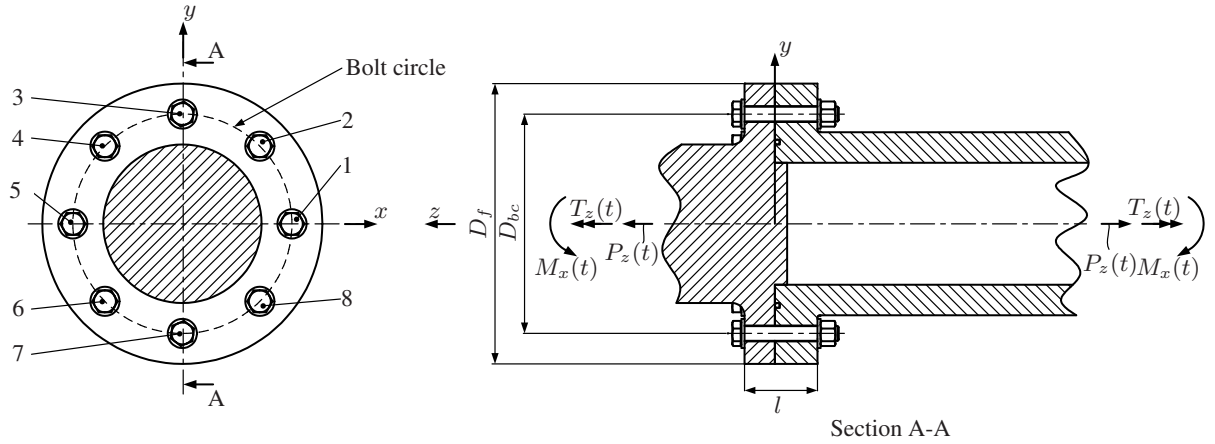


Figure 39: Pretensioned flange connection.

The diameter of the bolt circle is $D_{bc} = 180$ mm, i.e. the length of the bolt circle is:

$$l = \pi \cdot D_{bc} = \pi \cdot 180 \text{ mm} = 565.5 \text{ mm} \quad (264)$$

This means that the distance between the bolts measured on the bolt circle is

$$a_b = \frac{l}{8} = \frac{565.5 \text{ mm}}{8} = 71 \text{ mm} \quad (265)$$

This distance is approximately 6 times the bolt diameter $d = 12$ mm. This means that distance between the bolts is not too large but of course it could be smaller.

The distance between the bolt circle and the outer edge of the flange is:

$$a_f = \frac{1}{2} \cdot (D_f - D_{bc}) = \frac{1}{2} \cdot (230 \text{ mm} - 180 \text{ mm}) = 25 \text{ mm} > 2 \cdot d = 2 \cdot 12 \text{ mm} = 24 \text{ mm} \quad (266)$$

This means that the distance between the bolt circle and the outer edge of the flange is ok.

The ratio between the thickness l of the compressed material and the bolt diameter d is:

$$j = \frac{d}{l} = \frac{12 \text{ mm}}{60 \text{ mm}} = 0.20 \quad (267)$$

For pretensioned connections this ratio should be ≤ 0.25 which is also the case here.

Since the connection in question is permanent and not expected to be disassembled and assembled again within its lifetime the pretension force is chosen to be:

$$F_i = 0.90 \cdot S_p \cdot A_t \quad (268)$$

where

$$A_t = 84.27 \text{ mm}^2 \text{ (see Norton table 15-2)}$$

$$S_p = 600 \text{ MPa (see Norton table 15-7)}$$

This means that

$$F_i = 0.90 \cdot 600 \frac{\text{N}}{\text{mm}^2} \cdot 84.27 \text{ mm}^2 = 45510 \text{ N} \quad (269)$$

The surface pressure p under the hardened washers is computed as

$$p = \frac{F_i}{A_{washer}} = \frac{45510 \text{ N}}{\frac{\pi}{4} \cdot ((24 \text{ mm})^2 - (13 \text{ mm})^2)} = 142 \text{ MPa} < p_g = 420 \text{ MPa} \quad (270)$$

Since the surface pressure p is lower than the limiting surface pressure p_g then there is no reason to fear that the connection will come loose as result of permanent deformations in the material right under the nut and washer.

For oil lubricated threads the following relation between the pretension force F_i and the pretension moment T_i is known:

$$T_i \cong 0.21 \cdot F_i \cdot d \cong 0.21 \cdot 45510 \text{ N} \cdot 12 \text{ mm} \cong 1.15 \cdot 10^5 \text{ Nmm} \cong 115 \text{ Nm} \quad (271)$$

Determination of the bolt force under the assumption that the bolts were not pretensioned:

The normal force $P_z(t)$ is distributed equally on the 8 bolts and causes the following maximum tensile force in all bolts:

$$P_{normalforce,max} = \frac{P_{z,max}}{8} = \frac{32000 \text{ N}}{8} = 4000 \text{ N} \quad (272)$$

The bending moment $M_x(t)$ causes the following normal forces in the bolts:

$$P_{i,bending} = \frac{M_x(t) \cdot y_i}{\sum_{j=1}^8 y_j^2} \quad (273)$$

For the present configuration of the connection the following is found:

$$\begin{aligned} \sum_{j=1}^8 y_j^2 &= 2 \cdot 0^2 + 4 \cdot \left(\frac{D_{bc}}{2} \cdot \sin 45^\circ \right)^2 + 2 \cdot \left(\frac{D_{bc}}{2} \right)^2 \\ &= 0 + 4 \cdot \left(\frac{180 \text{ mm}}{2} \cdot \sin 45^\circ \right)^2 + 2 \cdot \left(\frac{180 \text{ mm}}{2} \right)^2 \\ &= 16200 \text{ mm}^2 + 16200 \text{ mm}^2 \\ &= 32400 \text{ mm}^2 \end{aligned} \quad (274)$$

From this the force in the bolt with the highest tensile load (bolt number 3) can be found to be:

$$\begin{aligned}
 P_{3,bending,max} &= \frac{M_x(t) \cdot \frac{D_{bc}}{2}}{\sum_{j=1}^8 y_j^2} \\
 &= \frac{M_{x,maks} \cdot \frac{180 \text{ mm}}{2}}{32400 \text{ mm}^2} \\
 &= \frac{7200 \cdot 10^3 \text{ Nmm} \cdot 90 \text{ mm}}{32400 \text{ mm}^2} \\
 &= 20000 \text{ N}
 \end{aligned} \tag{275}$$

The torque $T_z(t)$ causes an equal shear force on all 8 bolts. This is due to the fact that all 8 bolts have the same distance to the area center of the bolts. The center is located in origo of the xyz -coordinate system.

$$V_{torque} = \frac{T_z(t) \cdot \frac{D_{bc}}{2}}{\sum_{j=1}^8 r_j^2} \tag{276}$$

For the present bolt configuration one can find

$$\sum_{j=1}^8 r_j^2 = 8 \cdot \left(\frac{D_{bc}}{2} \right)^2 = 8 \cdot \left(\frac{180 \text{ mm}}{2} \right)^2 = 64800 \text{ mm}^2 \tag{277}$$

This means that

$$V_{torque} = \frac{T_z(t) \cdot \frac{180 \text{ mm}}{2}}{64800 \text{ mm}^2} = \frac{\pm 1330 \cdot 10^3 \text{ Nmm} \cdot 90 \text{ mm}}{64800 \text{ mm}^2} = \pm 1847 \text{ N} \tag{278}$$

This means that if the bolts are not pretensioned then the bolt with the highest load will see a maximum normal force of

$$P_{3,maks} = P_{normalforce,max} + P_{3,bending,max} = 4000 \text{ N} + 20000 \text{ N} = 24000 \text{ N} \tag{279}$$

and a shear force of

$$|V_3|_{max} = 1847 \text{ N} \tag{280}$$

On the other hand, if the bolts are pretensioned to the pretension force $F_i = 45510 \text{ N}$ computed earlier then the friction force between the compressed parts can hold the shear force. This means that the bolt will not feel the shear force at all. In addition the bolts will only feel a small part of the external load on the connection. This is shown in the following.

Since both the flanges and the bolts are made of steel then the stiffness constant C for each bolt area can be found directly via equation 15.19 in Norton:

$$C = C_r = p_3 \cdot r^3 + p_2 \cdot r^2 + p_1 \cdot r + p_0 \tag{281}$$

where $r = 1$ and $p_0 = 0.6118$, $p_1 = -1.1715$, $p_2 = 1.0875$, $p_3 = -0.3806$ since $j = 0.20$

this means that

$$C = C_r = -0.3806 \cdot 1^3 + 1.0875 \cdot 1^2 - 1.1715 \cdot 1 + 0.6118 = 0.1472 \quad (282)$$

After this the external load felt by the pretensioned bolt P_{b3} and the relaxation felt by the compressed material P_m can be computed:

$$P_{b3,max} = C \cdot P_{3,x} = 0.1472 \cdot 24000 \text{ N} = 3533 \text{ N} \quad (283)$$

and

$$P_{m3,max} = (1 - C) \cdot P_{3,max} = (1 - 0.1472) \cdot 24000 \text{ N} = 20467 \text{ N} \quad (284)$$

This means that when the bolts are pretensioned then the maximum tensile force in the bolt with the highest load is

$$F_{b3,max} = F_i + P_{b3,max} = 45510 \text{ N} + 3533 \text{ N} = 49043 \text{ N} \quad (285)$$

while the compressive force on the compressed material under this bolt is

$$F_{m3,min} = F_i - P_{m3,max} = 45510 \text{ N} - 20467 \text{ N} = 25043 \text{ N} \quad (286)$$

These forces are shown in the bolt diagram in Fig. 40.

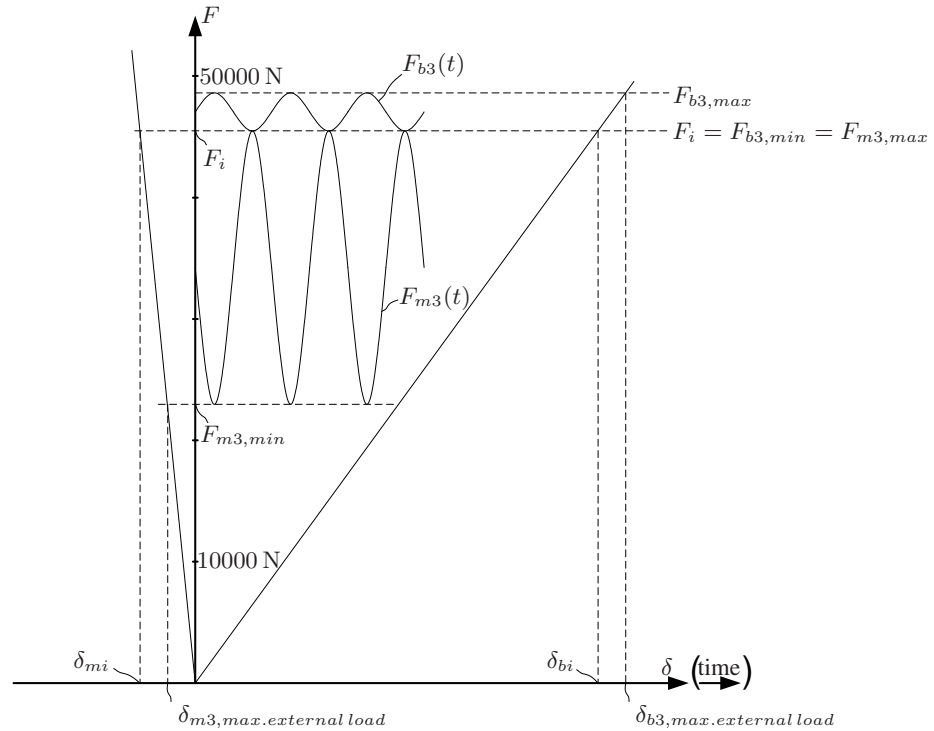


Figure 40: Bolt diagram for bolt number 3 in the flange connection shown in Fig. 39.

If the friction coefficient is assumed to $\mu \cong 0.15$ to be on the safe side then the connection under the bolt with the highest load will be able to transmit a friction force of at least

$$F_{friction,m3,min} = \mu \cdot F_{m3,min} = 0.15 \cdot 25043 \text{ N} = 3756 \text{ N} \quad (287)$$

Since $F_{friction,m3,min} > |V_3|_{max}$ then the friction force on the connected surfaces withstands the entire torque. This means that the bolts do not feel any shear forces at all in the correct pretensioned connection.

The safety factor against relative sliding between the connected surfaces can be computed as:

$$N_{slide} = \frac{F_{friction,m3,min}}{V_{3,max}} = \frac{3756 \text{ N}}{1847 \text{ N}} \approx 2.0 \quad (288)$$

The safety factor against separation of the connected surfaces N_{sep} is computed according to equation 15.14d in Norton:

$$N_{sep} = \frac{F_i}{P_{3,max} \cdot (1 - C)} = \frac{45510 \text{ N}}{24000 \text{ N} \cdot (1 - 0.1472)} \approx 2.2 \quad (289)$$

To be able to determine the safety factor against fatigue fracture then the nominal amplitude stress $\sigma_{a,nom}$ in the bolt with the highest load is computed:

$$\sigma_{a,nom} = \frac{F_{b3,max} - F_{b3,min}}{2 \cdot A_t} = \frac{F_{b3,max} - F_i}{2 \cdot A_t} = \frac{49043 \text{ N} - 45510 \text{ N}}{2 \cdot 84.27 \text{ mm}^2} = 21 \text{ MPa} \quad (290)$$

The fatigue limit that in common practice is considered to be independent of the mean stress in the bolt is computed for a reliability of approximately 50 %:

$$S_{e,50\%,galv} \cong C_{galv} \cdot \left\{ \frac{150 \text{ mm}}{d} + 45 \right\} \frac{\text{N}}{\text{mm}^2} \text{ where the bolt diameter } d \text{ is inserted in mm, ref. VDI 2230}$$

$$S_{e,50\%,galv} \cong 0.80 \cdot \left\{ \frac{150 \text{ mm}}{12 \text{ mm}} + 45 \right\} \frac{\text{N}}{\text{mm}^2} = 46 \frac{\text{N}}{\text{mm}^2} \quad (291)$$

This fatigue strength is converted to a reliability of 99.999 %

$$S_{e,99.999\%} \cong C_{reliab} \cdot S_{e,50\%,galv} = 0.659 \cdot 46 \text{ MPa} = 30 \text{ MPa} \quad (292)$$

$$C_{reliab} \cong 0.659 \text{ Norton table 6-4} \quad (293)$$

This means that the safety factor with a reliability of 99.999 % against fatigue fracture in the pretensioned bolt with highest load is:

$$N_{fat} = \frac{S_{e,99.999\%}}{\sigma_{a,nom}} = \frac{30 \text{ MPa}}{21 \text{ MPa}} = 1.4 \quad (294)$$

With the high reliability required in this exercise then it would have been sufficient with a safety factor $N_{fat} \geq 1.0$.

WRONG

Exercise OØM-MKE-18 solution

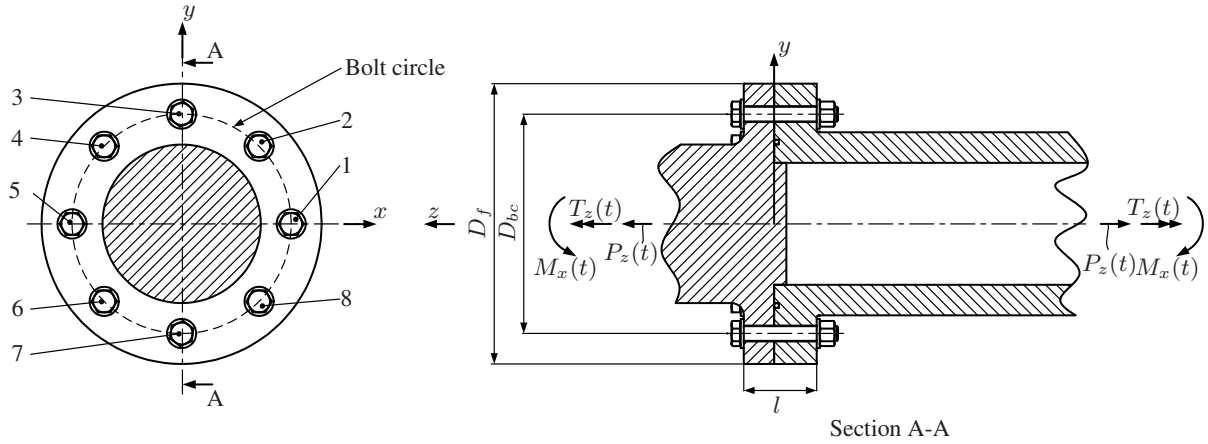


Figure 1: Pretensioned flange connection.

Control calculations for the bolt joint.

$$l = \pi \cdot D_{bc} = \pi \cdot 180 \text{ mm} = 565.5 \text{ mm} \quad (1)$$

$$a_b = \frac{l}{8} = \frac{565.5 \text{ mm}}{8} = 71 \text{ mm} \quad (2)$$

$$a_f = \frac{1}{2} \cdot (D_f - D_{bc}) = \frac{1}{2} \cdot (230 \text{ mm} - 180 \text{ mm}) = 25 \text{ mm} > 2 \cdot d = 2 \cdot 12 \text{ mm} = 24 \text{ mm} \quad (3)$$

$$j = \frac{d}{l} = \frac{12 \text{ mm}}{60 \text{ mm}} = 0.20 \quad (4)$$

$$F_i = 0.90 \cdot S_p \cdot A_t \quad (5)$$

$$A_t = 84.27 \text{ mm}^2 \text{ (see Norton table 15-2)}$$

$$S_p = 600 \text{ MPa (see Norton table 15-7)}$$

$$F_i = 0.90 \cdot 600 \frac{\text{N}}{\text{mm}^2} \cdot 84.27 \text{ mm}^2 = 45510 \text{ N} \quad (6)$$

$$p = \frac{F_i}{A_{washer}} = \frac{45510 \text{ N}}{\frac{\pi}{4} \cdot ((24 \text{ mm})^2 - (13 \text{ mm})^2)} = 142 \text{ MPa} < p_g = 420 \text{ MPa} \quad (7)$$

$$T_i \cong 0.21 \cdot F_i \cdot d \cong 0.21 \cdot 45510 \text{ N} \cdot 12 \text{ mm} \cong 1.15 \cdot 10^5 \text{ Nmm} \cong 115 \text{ Nm} \quad (8)$$

$$P_{normalforce,max} = \frac{P_{z,maks}}{8} = \frac{32000 \text{ N}}{8} = 4000 \text{ N} \quad (9)$$

$$P_{i,bending} = \frac{M_x(t) \cdot y_i}{\sum_{j=1}^8 y_j^2} \quad (10)$$

WRONG

WRONG

$$\begin{aligned}\sum_{j=1}^8 y_j^2 &= 2 \cdot 0^2 + 4 \cdot \left(\frac{D_{bc}}{2} \cdot \sin 45^\circ \right)^2 + 2 \cdot \left(\frac{D_{bc}}{2} \right)^2 \\ &= 0 + 4 \cdot \left(\frac{180 \text{ mm}}{2} \cdot \sin 45^\circ \right)^2 + 2 \cdot \left(\frac{180 \text{ mm}}{2} \right)^2 \\ &= 16200 \text{ mm}^2 + 16200 \text{ mm}^2 \\ &= 32400 \text{ mm}^2\end{aligned}\tag{11}$$

$$\begin{aligned}P_{3,bending,max} &= \frac{M_x(t) \cdot \frac{D_{bc}}{2}}{\sum_{j=1}^8 y_j^2} \\ &= \frac{M_{x,maks} \cdot \frac{180 \text{ mm}}{2}}{32400 \text{ mm}^2} \\ &= \frac{7200 \cdot 10^3 \text{ Nmm} \cdot 90 \text{ mm}}{32400 \text{ mm}^2} \\ &= 20000 \text{ N}\end{aligned}\tag{12}$$

$$V_{torque} = \frac{T_z(t) \cdot \frac{D_{bc}}{2}}{\sum_{j=1}^8 r_j^2}\tag{13}$$

$$\sum_{j=1}^8 r_j^2 = 8 \cdot \left(\frac{D_{bc}}{2} \right)^2 = 8 \cdot \left(\frac{180 \text{ mm}}{2} \right)^2 = 64800 \text{ mm}^2\tag{14}$$

$$V_{torque} = \frac{T_z(t) \cdot \frac{180 \text{ mm}}{2}}{64800 \text{ mm}^2} = \frac{\pm 1330 \cdot 10^3 \text{ Nmm} \cdot 90 \text{ mm}}{64800 \text{ mm}^2} = \pm 1847 \text{ N}\tag{15}$$

$$P_{3,maks} = P_{normalforce,max} + P_{3,bending,max} = 4000 \text{ N} + 20000 \text{ N} = 24000 \text{ N}\tag{16}$$

$$|V_3|_{max} = 1847 \text{ N}\tag{17}$$

$$C = C_r = p_3 \cdot r^3 + p_2 \cdot r^2 + p_1 \cdot r + p_0\tag{18}$$

where $r = 1$ and $p_0 = 0.6118$, $p_1 = -1.1715$, $p_2 = 1.0875$, $p_3 = -0.3806$ since $j = 0.20$

$$C = C_r = -0.3806 \cdot 1^3 + 1.0875 \cdot 1^2 - 1.1715 \cdot 1 + 0.6118 = 0.1472\tag{19}$$

$$P_{b3,max} = C \cdot P_{3,x} = 0.1472 \cdot 24000 \text{ N} = 3533 \text{ N}\tag{20}$$

$$P_{m3,max} = (1 - C) \cdot P_{3,max} = (1 - 0.1472) \cdot 24000 \text{ N} = 20467 \text{ N}\tag{21}$$

$$F_{b3,max} = F_i + P_{b3,max} = 45510 \text{ N} + 3533 \text{ N} = 49043 \text{ N}\tag{22}$$

$$F_{m3,min} = F_i - P_{m3,max} = 45510 \text{ N} - 20467 \text{ N} = 25043 \text{ N}\tag{23}$$

WRONG

WRONG

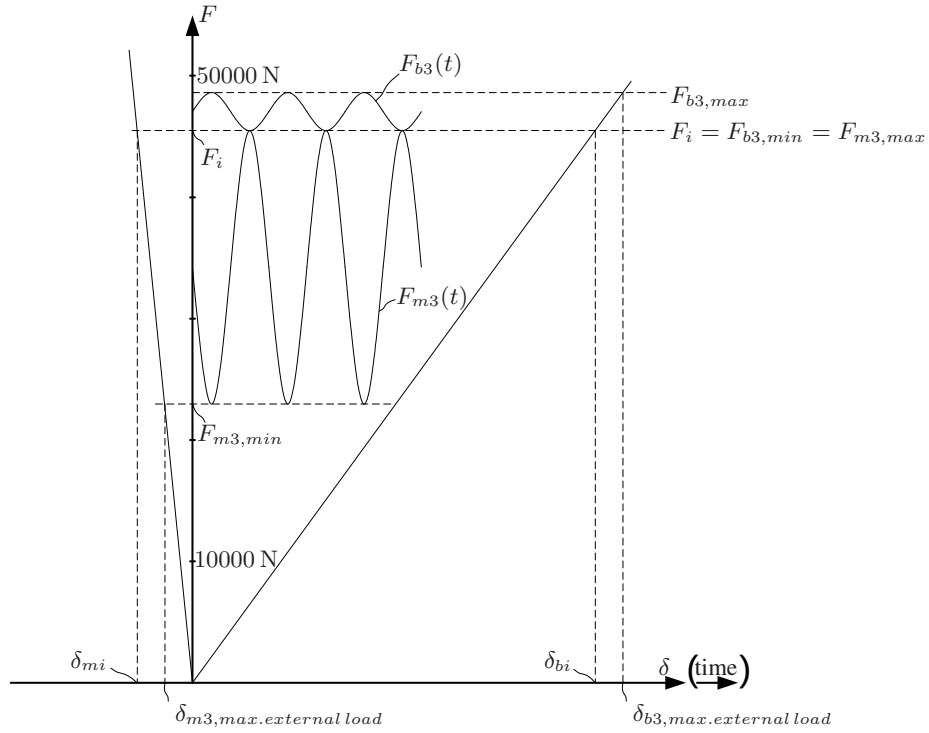


Figure 2: Bolt diagram for bolt number 3 in the flange connection shown in figure 1.

$$F_{friction,m3,min} = \mu \cdot F_{m3,min} = 0.15 \cdot 25043 \text{ N} = 3756 \text{ N} \quad (24)$$

$$N_{slide} = \frac{F_{friction,m3,min}}{V_{3,max}} = \frac{3756 \text{ N}}{1847 \text{ N}} \approx 2.0 \quad (25)$$

$$N_{sep} = \frac{F_i}{P_{3,max} \cdot (1 - C)} = \frac{45510 \text{ N}}{24000 \text{ N} \cdot (1 - 0.1472)} \approx 2.2 \quad (26)$$

$$\sigma_{a,nom} = \frac{F_{b3,max} - F_{b3,min}}{2 \cdot A_t} = \frac{F_{b3,max} - F_i}{2 \cdot A_t} = \frac{49043 \text{ N} - 45510 \text{ N}}{2 \cdot 84.27 \text{ mm}^2} = 21 \text{ MPa} \quad (27)$$

$$S_{e,50\%,galv} \cong C_{galv} \cdot \left\{ \frac{150 \text{ mm}}{d} + 45 \right\} \frac{\text{N}}{\text{mm}^2} \text{ where the bolt diameter } d \text{ is inserted in mm, ref. VDI 2230}$$

$$S_{e,50\%,galv} \cong 0.80 \cdot \left\{ \frac{150 \text{ mm}}{12 \text{ mm}} + 45 \right\} \frac{\text{N}}{\text{mm}^2} = 46 \frac{\text{N}}{\text{mm}^2} \quad (28)$$

$$S_{e,99.999\%} \cong C_{reliab} \cdot S_{e,50\%,galv} = 0.659 \cdot 46 \text{ MPa} = 30 \text{ MPa} \quad (29)$$

$$C_{reliab} \cong 0.659 \text{ Norton table 6-4} \quad (30)$$

$$N_{fat} = \frac{S_{e,99.999\%}}{\sigma_{a,nom}} = \frac{30 \text{ MPa}}{21 \text{ MPa}} = 1.4 \quad (31)$$

WRONG

General about figures

- Aim for equal quality of the figures within the different sections.
- Good sketches done by hand can be OK (purpose and time?).
- More accurate figures: use the computer.
- The figures must be used in the text.
- Figures are easier to understand than text.
- Figures can replace text.
- If you need a figure/sketch when writing, then the reader needs it when reading.
- Programs?
- Filetypes?
- Good figures take time to make - **ACCEPT IT**

Case A

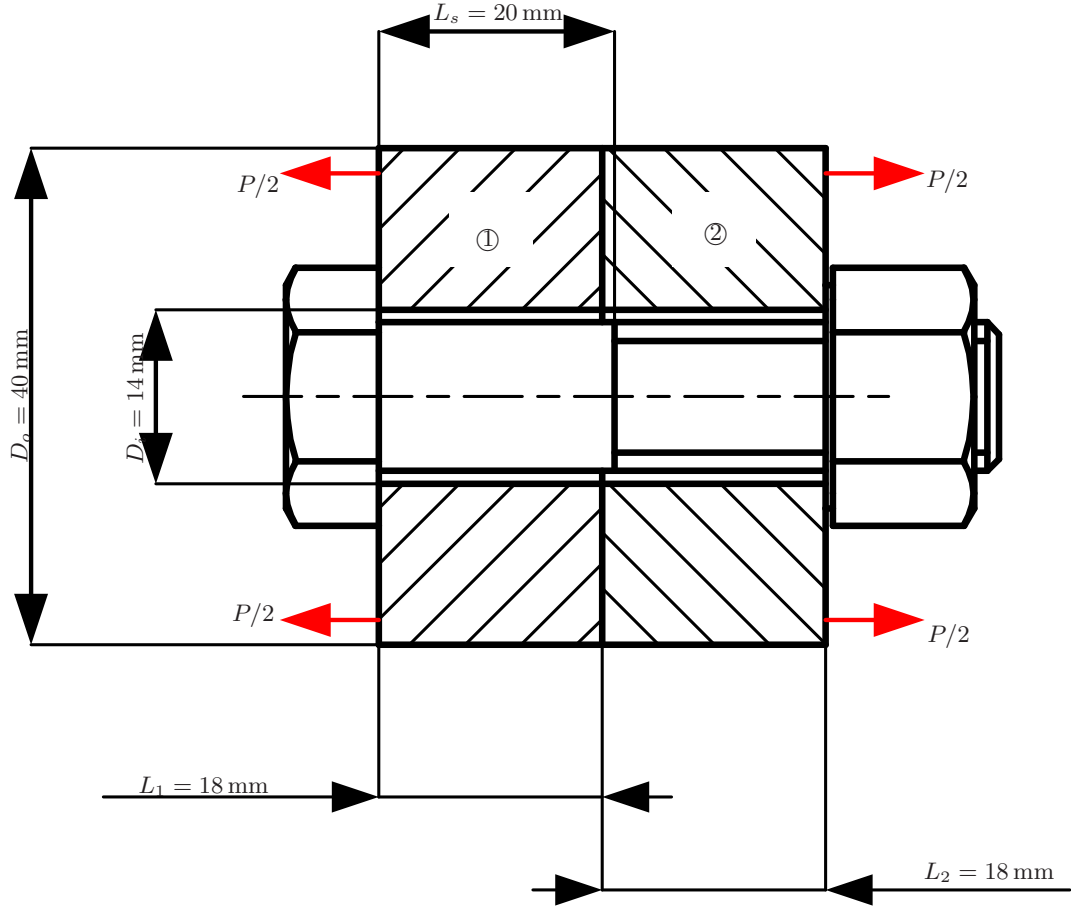


Figure 1: Four variants of a preloaded bolt connection with an external load P .

The stiffness of the steel discs ① and ② are computed. It is used that $L_1 = L_2$.

$$\begin{aligned}
 k_1 = k_2 &= \frac{E \cdot A}{L_1} \\
 &= \frac{E \cdot \frac{(D_o^2 - D_i^2) \cdot \pi}{4}}{L_1} \\
 &= \frac{205 \cdot 10^3 \text{ MPa} \cdot \frac{((40 \text{ mm})^2 - (14 \text{ mm})^2) \cdot \pi}{4}}{18 \text{ mm}} \\
 &= 12558517 \frac{\text{N}}{\text{mm}}
 \end{aligned} \tag{1}$$

The tensile stress area of a M12 fine thread bolt is $A_t = 92.07 \text{ mm}^2$, see table 15-2 in [1], and the minimum proof strength of grade 12.9 is $S_P = 970 \text{ MPa}$, se table 15-7 in [1]. Thereby the preload of the bolt and the compressed material is computed as:

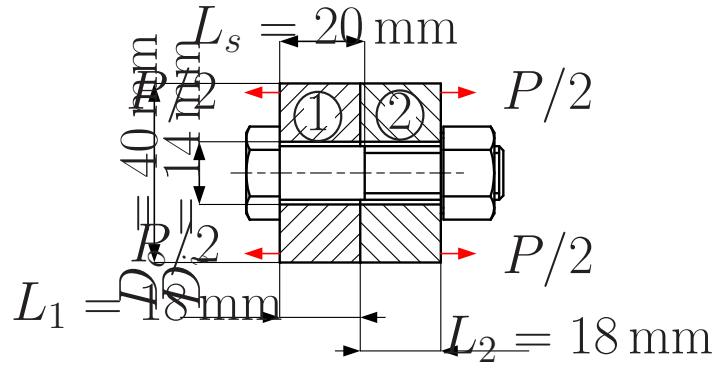


Figure 2: Four variants of a preloaded bolt connection with an external load P .

The stiffness of the steel discs ① and ② are computed. It is used that $L_1 = L_2$.

$$\begin{aligned}
 k_1 = k_2 &= \frac{E \cdot A}{L_1} \\
 &= \frac{E \cdot \frac{(D_o^2 - D_i^2) \cdot \pi}{4}}{L_1} \\
 &= \frac{205 \cdot 10^3 \text{ MPa} \cdot \frac{((40 \text{ mm})^2 - (14 \text{ mm})^2) \cdot \pi}{4}}{18 \text{ mm}} \\
 &= 12558517 \frac{\text{N}}{\text{mm}}
 \end{aligned} \tag{2}$$

The tensile stress area of a M12 fine thread bolt is $A_t = 92.07 \text{ mm}^2$, see table 15-2 in [1], and the minimum proof strength of grade 12.9 is $S_P = 970 \text{ MPa}$, se table 15-7 in [1]. Thereby the preload of the bolt and the compressed material is computed as:

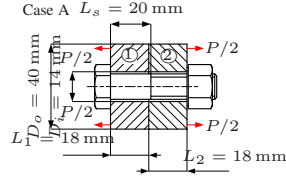


Figure 3: Four variants of a preloaded bolt connection with an external load P .

The stiffness of the steel discs ① and ② are computed. It is used that $L_1 = L_2$.

$$\begin{aligned}
 k_1 = k_2 &= \frac{E \cdot A}{L_1} \\
 &= \frac{E \cdot \frac{(D_O^2 - D_i^2) \cdot \pi}{4}}{L_1} \\
 &= \frac{205 \cdot 10^3 \text{ MPa} \cdot \frac{((40 \text{ mm})^2 - (14 \text{ mm})^2) \cdot \pi}{4}}{18 \text{ mm}} \\
 &= 12558517 \frac{\text{N}}{\text{mm}}
 \end{aligned} \tag{3}$$

The tensile stress area of a M12 fine thread bolt is $A_t = 92.07 \text{ mm}^2$, see table 15-2 in [1], and the minimum proof strength of grade 12.9 is $S_P = 970 \text{ MPa}$, see table 15-7 in [1]. Thereby the preload of the bolt and the compressed material is computed as:

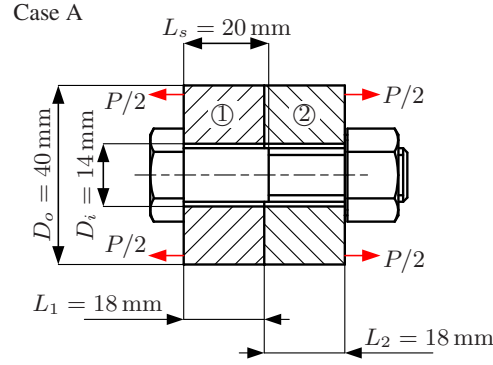


Figure 4: Four variants of a preloaded bolt connection with an external load P .

The stiffness of the steel discs ① and ② are computed. It is used that $L_1 = L_2$.

$$\begin{aligned}
 k_1 = k_2 &= \frac{E \cdot A}{L_1} \\
 &= \frac{E \cdot \frac{(D_o^2 - D_i^2) \cdot \pi}{4}}{L_1} \\
 &= \frac{205 \cdot 10^3 \text{ MPa} \cdot \frac{((40 \text{ mm})^2 - (14 \text{ mm})^2) \cdot \pi}{4}}{18 \text{ mm}} \\
 &= 12558517 \frac{\text{N}}{\text{mm}}
 \end{aligned} \tag{4}$$

The tensile stress area of a M12 fine thread bolt is $A_t = 92.07 \text{ mm}^2$, see table 15-2 in [1], and the minimum proof strength of grade 12.9 is $S_P = 970 \text{ MPa}$, see table 15-7 in [1]. Thereby the preload of the bolt and the compressed material is computed as:

Graphs

- Excel, Matlab.....
- Quality
 - Lines must be visible.
 - Text must be readable (font?)
 - Clear.
 - Designations on the axes.
 - Units.
 - Graphs for comparison: same divisions on the axes.
- You are responsible that it is readable

Commenting and proof reading #1

- Create a routine and culture where it is accepted to comment on a text.
- It is difficult to check your own text - but it can be learned through training.
- If possible get a colleague to check the text.
- Read the text two times
 - One time for the overall structure and content - red line through the text.
 - One time for details, - see next slide
- Read through the text and write comments. Do the corrections afterwards so you do not loose concentration during reading.

Commenting and proof reading #2

Things to consider when proof reading:

- Singular/plural form of the words
- Past tense/present tense
- Spelling
- Does each sentence make sense? Are any words missing? (Read what is written NOT what you think is written). Is the meaning clear?
- Are the sentences too long? > shorten/rewrite
- Do the sentences make correct connection to the previous sentence and the next sentence?

Commenting and proof reading #3

Things to consider when proof reading:

- Avoid spoken language like "don't", "doesn't" and "can't". In written language it is "do not", "does not" and "cannot(/can not)"
- Figure text and number below all figures. All figures must be referenced in the text.
- Table text and number above all tables. All tables must be referenced in the text.
- Are abbreviations defined/explained before they are used?
- Consistent use of multiplication sign (if used)
- Are the results and graphs explained? The author must draw the conclusion NOT the reader.

Commenting and proof reading #4

Things to consider when proof reading:

- Are the figures clear and understandable? Type of graphics file.
- Does the text on figures have a reasonable font size? Usually, it is appropriate with a font size about the same as the figure text.
- Is the formatting of variables on figures similar to the formatting in the equations and text? It is recommended.
- Etc.